Proceedings of the Design Society, Volume 6: DESIGN 2026
https://doi.org/10.1017/pds.2026.10438

Rapid prototyping PCBs: recommendations for utilising PCB production as a design tool

Jonas Rossavik[1,✉], Håvard Vestad[1], Carlo Kriesi[2], Martin Francis Berg[1] and Martin Steinert[1]

[1] Norwegian University of Science and Technology, Norway, [2] Vitroscope AS, Norway

✉ jonas.rossavik@ntnu.no

ABSTRACT: Printed circuit boards (PCBs) fix and connect electrical components and are widely used. Current design methods emphasise mature products and do not leverage the potential of PCBs as prototyping tools. Accordingly, an alternative approach using PCBs for prototyping electrical and mechatronic solutions is evaluated through three case studies. Insights formed five concrete recommendations for designers: Increase fidelity deliberately, design for prototyping, iterate incrementally, parallelise prototyping, and prototype and test early. These aim to make prototyping with PCBs more accessible.

KEYWORDS: rapid prototyping, case study, design tools, design methodology

1. Introduction

1.1. Printed Circuit Boards

Printed Circuit Boards (PCB) are a backbone of the modern electronics industry (Canal Marques et al., 2013; LaDou, 2006). Conceptually, a PCB is a plate of conductors that can electrically connect and mechanically fix electronic components; They provide a robust mechanical and electrical foundation for electrical systems. The global PCB market reached a size of USD 73.1 Billion in 2024, and is projected to grow (Research and Markets, 2025). They are increasingly complex, such as multi-layer PCBs, flexible printed circuits (FPC), or flexi-rigid PCBs; offer fine resolution; and benefit from low costs (LaDou, 2006; Moreira et al., 2016; Moschou & Tserepi, 2017; Shamkhalichenar et al., 2020). These factors, in addition to increased availability of low-volume PCB production (Moschou & Tserepi, 2017), imply PCBs are a powerful tool in development projects with electronics.

PCBs are increasingly integrated in the development of concepts and products. In recent decades, there has been a substantial rise of interdisciplinarity in product development (Berschik & Krause, 2024; Bilal et al., 2025). PCBs offer many advantages over traditional breadboards, and certain prototype requirements make their use necessary, such as space-efficiency, signal integrity, robustness, or repeatability. The capabilities of PCBs also enable them to be used in alternative ways, constructing sensors (Moreira et al., 2016; Shamkhalichenar et al., 2020), antennas (Low et al., 2005) heating elements (Vu et al., 2024), amongst several other alternative uses for PCBs (Han et al., 2015; Jeong et al., 2018; Perdigones, 2021). Despite this, there are limitations holding back the full potential of PCBs in the context of rapid prototyping, namely time, machinery and skill. Whether one wants or needs to prototype with PCBs, key questions remain: When should you prototype with PCBs, and how do you rapidly prototype with PCBs? As no clear guidelines exist that answer these questions, designers are largely on their own in deciding approach and method.

1.2. The PCB design process

Literature on the PCB design process is limited, with few recent contributions, and no Design for PCB frameworks. The closest we get is Design for Manufacturability models which optimize cost and

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

performance of PCB assembly (Rodgers, 2016). The *Global Electronics Association*, formerly *IPC (Institute of Printed Circuits)*, provides a generic standard for PCB design, namely *IPC-2221: Generic Standard on Printed Board Design* (IPC, 2003). The purpose of the standard is as follows:

> *The requirements contained herein are intended to establish design principles and recommendations that shall be used in conjunction with the detailed requirements of a specific interconnecting structure sectional standard to produce detailed designs intended to mount and attach passive and active components.*

While providing a substantial amount design requirements and rules with regards to materials, conductor clearances, documentation, it does not lay out a detailed workflow for PCB design. The closest it gets is a flow chart of the PCB design and fabrication sequence. PCB design involves "Part list", "Schematic and logic diagram", "End product requirements", "PCB layout", and "Documentation", and follows a sequential process. The *Printed Circuits Handbook* lays out a more detailed typical hardware prototyping design process (Ritchey, 2016), represented in Figure 1. The process follows a highly linear manner, before opening for iterating is presented in the final step. Testing the PCB does not occur until the 10th process step. The author notes that there are typically several, sometimes many, iterations of the PCB to isolate defects. They also add that it is often a time-consuming process and go on to suggest a "virtual" prototyping design flow, focusing on simulation of designs that are entirely digital, thus enabling rapid iteration. Though this is deployed in the development of high performance computers and complicated microchips, it is not universally used in PCB design (Ritchey, 2016).

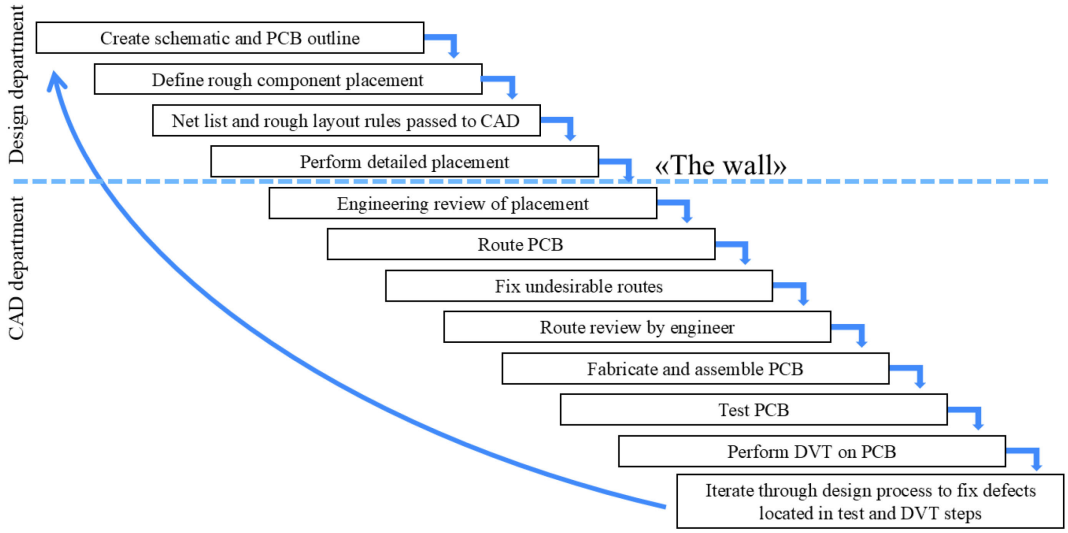


Figure 1. A typical hardware prototyping design process. Content and handovers are adapted from Ritchey (2016), but plotted as a step down process to highlight its similarity to other linear sequential product development processes

To the authors' knowledge there are no published methods on rapid prototyping of PCBs. There is a high likelihood that established PCB design methods are effective for developing circuitry of high complexity, but what about PCB designs where the list of requirements is not fixed, how do you proceed then? Existing frameworks do not open for ambiguous problem and solution spaces with dynamic requirements where iteration cycles of building and testing are employed early in the development process (Kriesi et al., 2016).

1.3. On rapid prototyping in a PCB context

Rapid prototyping entails performing fast iteration cycles through prototyping and can increase the rate of learning (Leifer & Steinert, 2011) and yield more valuable design insights than spending that time on a single iteration (Dow et al., 2009). Iterative processes are also preferred as they obtain better design outcomes (Ulrich & Eppinger, 2016). Time is naturally of high importance when seeking fast iteration cycles. One should see data early and iterate based on it. Extending the prototyping phase can yield better decisions when developing complex interdisciplinary concepts (Sjöman, 2019).

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

It is necessary to define some key terms for clarification. Resolution is the amount of detail of a prototype, while fidelity is its closeness to the eventual design (Houde & Hill, 1997).
In the development of mechanical products and concepts, rapid prototyping has been enabled by technologies such as laser cutters and 3D printers (Zivanovic et al., 2020), which have enabled prototyping of much higher fidelity and resolution, at an earlier stage in the development process. A similar shift towards desktop machines for production is also present in PCB design (Yan, 2024), however their use is not extensive as of yet. There is a growing interest in additive PCB manufacturing methods (Goti, 2025b), however the market is currently relatively small (Espera et al., 2019). This is arguably due to relatively high cost and low availability of PCB manufacturing devices, especially when compared to the low cost and high availability of outsourcing production (Moreira et al., 2016; Moschou & Tserepi, 2017; Shamkhalichenar et al., 2020).
Several factors hinder the utilisation of outsourced PCB manufacturing in a rapid prototyping context, however. PCB design can be time-consuming (Yan et al., 2022) which slows iteration cycles and increases the risk of design fixation through sunk-cost fallacy (Jansson & Smith, 1991; Youmans, 2011). Lead times for manufacturing, shipping, and customs further decelerate the iteration cycle. There is a skill gap, as PCB prototyping requires skills with often unfamiliar machinery and design software (H. Vestad et al., 2019). Typically complex in nature, the difficulty in fixing mistakes and iterating designs is a major limitation of prototyping with PCBs (Narumi et al., 2015), and there is a need for easy redesign (Goti, 2025a). Simplified, complex PCBs either work or they don't, and the possibility of physical modifications of a manufactured PCB is limited.

1.4. Aims and methods

This article explores the usage of rapid prototyping principles in a PCB prototyping context and aims to enable PCB manufacturing methods to be used as a rapid prototyping tool. We applied documented rapid prototyping to development processes involving PCBs to investigate its effect. Through three case studies we tested rapid prototyping principles in a PCB design context, and gathered and compared insights to form hypotheses (Eisenhardt, 1989). Insights were generalized to form five recommendations for enabling rapid prototyping of PCBs based on experience from the case studies. These guidelines aim to provide answers to when you should prototype with PCBs, and how to rapidly prototype with PCBs.

2. Cases

Case studies performed included two sensor development cases where PCBs were used as a prototyping tool, and one where a PCB heating element was developed. Sections present the case, the development process, and subsequent insights.

2.1. Case 1: pressure sensitive mat

We set out to prototype a pressure sensitive mat that estimates the location of chest compressions during simulated cardiopulmonary resuscitation. The core sensing principle is a resistive matrix: two orthogonal electrode sets on opposing sides of a piezoresistive sheet. Throughout the process it was deemed necessary to move to PCB production as a prototyping tool to obtain the required resolution. The process is laid out in Figure 2.

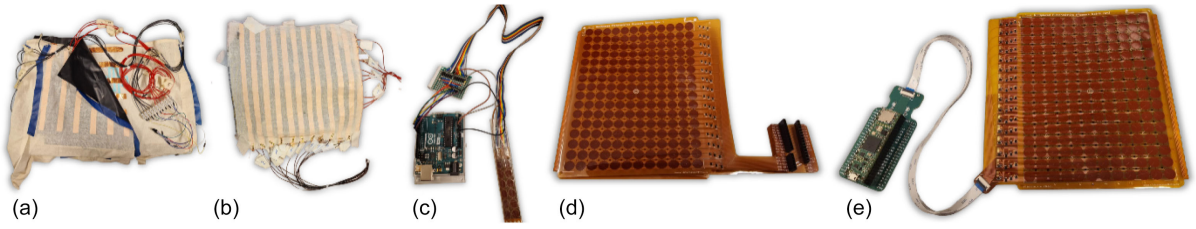


Figure 2. Key prototypes of pressure sensitive mat; (a) concept prototype; (b) finer pitch hand build; (c) small-scale FPC; (d) first full-scale FPC; (e) final prototype

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

2.1.1. Process

2(a): Concept and baseline. The concept was validated rapidly, as shown in Figure 2(b), using tote-bag fabric, 10 mm copper-tape strips, and Velostat as the piezoresistive material between layers. All materials were available in the lab. Paper fasteners and jumper leads connected strips to a breadboarded microcontroller. This prototype validated the principle and enabled fast iteration of readout circuitry and firmware, but exposed limitations in sensor resolution from high electrode pitch.
2(b): Finer pitch hand build. A second hand-built prototype retained bar electrodes while reducing pitch to test if it would improve sensor resolution. Circuitry and firmware were further developed, introducing diodes for each driven electrode. Testing with a CNC-driven rig showed improved accuracy but highlighted low mechanical robustness through loose wires and tape detachment. Further refinement of sensor resolution was also required.
2(c): Small-scale FPC. We outsourced fabrication of a 1x6 FPC tile consisting of two slim boards (rows, six cells with individual solder pads, and column, six connected cells with one common solder pad). Jumper leads were connected to solder pads and were once again connected to a breadboarded microcontroller. This enabled parallel iteration on circuitry. Testing indicated much improved performance. Each driven cell (rows) being individually wired enabled the evaluation of one diode per driven cell instead of per driven row, which marked reduced inter-cell bleed, further improving performance. Two limitations surfaced however, flying leads detached solder pads, and alignment of the rows and column layers was cumbersome.
2(d): First full-scale FPC. With architecture confirmed, we optimistically scaled to 12x16 cells on a single panel. The design integrated per-cell diodes, both layers on one panel with a designed fold line for alignment, and an integrated flex tail with female header pins for the microcontroller. This iteration revealed extensive weaknesses: a wiring error, strain and disconnection around header pins, and polyimide curl after assembly that decreased flatness.
2(e): Final prototype. The next revision retained the designed fold but added a solder bridge locking mechanism around the panel to retain flatness with minimal added thickness. We replaced the integrated flex tail with board-mounted flat FPC connectors and an off-the-shelf FPC cable to a small PCB interface board hosting the microcontroller. While the board was in fabrication, firmware development continued using the previous assembly. The resulting prototype had the resolution, robustness, and repeatability required for extended testing.

2.1.2. Insights

PCB production became necessary once resolution needs exceeded hand fabrication limits. This was discovered early due to rapid hand fabricated prototypes. Modularity in the form of circuitry being breadboarded through most prototypes, enabled rapid iteration on circuitry. A small scale FPC prototype was pivotal in gaining insights that informed proceeding produced prototypes, with the following full-scale FPC prototype also informing several design choices. Parallel iteration on mechanical, electrical, and software design shortened the effective iteration cycle despite fabrication lead times.
Case 1 also gave insights into possible improvements the process would have benefitted from. Prototype 2(e) prematurely scaled to a full-scale prototype, implementing new geometry, circuitry, components, microcontroller connectivity, and alignment procedures. The prototype required several days of design work, and while answering several design questions, identifying and separating these inquiries into individual prototypes with clear goals would have saved substantial design time. These could even have been manufactured on a panelised FPC.

2.2. Hair cell flow sensors

To orient themselves in water and feel flows around their bodies, fish use hair cells which are moved by the flows over their skin. To mimic this effect on a macro level, we proposed a simple design where a conductive hair is moved between pads, to detect alternating vortex streets in moving waters. The hair and the water in which it moves then acts as a variable resistor and the position of the hair and its collective distance(s) to the pad can be read as a changing voltage across a voltage divider, see Figure 3 (a). Section is adapted from Vestad, 2018.

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press



Figure 3. Key prototypes of hair cell flow sensors; (a) schematic of concept; (b) laser-cut large-scale prototype (c) simple prototype board prototype and prototype board with 2x2 sensor array (d) milled PCB with four sensors

2.2.1. Process

The concept was hypothesized as sketched in Figure 3a. To test whether the principle would work, a large-scale prototype model was mocked up as seen in Figure 3b. The model was made by gluing carbon fibres to electrical leads, and making pads from laser cut pieces covered in copper tape. The four pads were connected to LEDs, which would light up and change brightness dependent on the conductivity between pads and carbon fibres. Though the position of the hair would influence the brightness of the LEDs, the many hairs and large scale made it hard to gauge whether there were significant signal changes relating to changes in flow conditions with the prototype.

Smaller scale models were mocked up, Figure 3c, using prototype boards. Four pads were soldered to signal wires, while some carbon fibre hairs were mechanically, and electrically, fixed in the centre pad with solder. The smaller scale allowed for also making and fitting an array of four sensors in our test-setup to see differences between the measured points. This was the highest density of sensors we were able to fit by manually making these sensors on prototype boards within a reasonable amount of effort.

To increase the density of sensors even further, to target their placement in the alternating flow conditions, we moved into manufacturing of PCBs. As the way the PCB is being used as a hair cell sensor requires some less traditional design rules for the PCB, and as this was part of a fast explorative prototyping process, it was decided to use an in-house mill to mill out the PCB. This adds some extra steps on the designer's end, once the PCB design is complete a tool path needs to be generated as well as post processing for the CNC, but though the effort is higher for the designer - the PCB was made the same day, effectively allowing for the design-build-test cycle to continue.

2.2.2. Insights

Making fast models with available materials enabled the designer to gauge which considerations were of importance before drawing up a design for manufacturing. Eventually, the need for repeatability and smaller scale/higher density meant that PCB production was needed for further prototyping. Producing the PCBs with in-house equipment might demand more investment in effort from the developer, but the lead time can be reduced drastically so that the iterative prototyping process can continue.

As the long hairs resulted in slow positional changes, the sensors concept, as prototyped, were not efficient in detecting fast flow changes which were of interest for our application. Instead, materials with piezo-resistive properties discovered through this prototyping journey were pursued (Vestad & Steinert, 2019). This pivotal change in direction was due to serendipitous findings found through the iterative prototyping journey.

2.3. Case 3: FPC nickel-chrome heater

Case 3 follows the development of a PCB heating element. There was a need for a compact heater with low thermal inertia. Commercial ultra-thin heaters were unavailable at prototype quantities; thus, we explored in-house fabrication using nickel-chrome (NiCr) foil. Key prototypes from the development process are found in Figure 4. The section is adapted from Kriesi, 2018.

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press



Figure 4. Key prototypes of FPC NiCr heater; (a) available space 3D-print; (b) paper prototype; (c) laser-cut NiCr on tape; (d) full-scale silicone and NiCr heater; (e) manufactured FPC heater

2.3.1. Process

4(a): 3D-print of available space. The prototype framed the constraints and invited ideas for heating element designs.
4(b): Paper prototype. The first solution prototype was made to communicate the idea of using a thick-film style heaters with long trace patterns, as they have high shape flexibility and can be made very thin.
4(c): Laser-cut NiCr on tape. Due to difficulty of acquiring commercial ultra-thin heaters, and the desire for rapid prototypes, it was decided to fabricate prototypes in-house. Apparent materials and machinery were a 0.01 mm foil of NiCr (with temperature independent electrical resistance) and a 100 W CO2 laser cutter. Expertise found online discouraged the use of CO2-lasers for metal cutting, and clearly stated that it would never work, but it was attempted nevertheless - and worked. A small-scale heating element was laser cut and wrapped in a piece of clear adhesive tape, yielding a simple, flexible, and watertight heater to test early in the process. Connecting it to a battery showed it could heat but uncovered that the outer material would curl from heat and needed to be replaced.
4(d): Full-scale silicone and NiCr prototype. Following the concept validation, the next steps involved several theoretical iterations. These showed that it was indeed possible to obtain the amount of thermal energy required within the build volume and without too complex production methods. This information was used to design a NiCr heater with a fitting resistance, aiming for a homogeneous heat distribution. The heater would be sandwiched between sheets of silicone foil for increased heat resistance of the outer layer. At only 0.01 mm thickness, the NiCr was prone to fold and wrinkle, necessitating testing of production protocols. We landed on building a vacuum table that would fit inside of the laser cutter. Final silicone and NiCr heater met requirements and was even used in a commercialized product (Kriesi et al., 2019).
4(e): Manufactured FPC heater. Insight gathered from laser-cut NiCr enabled confident outsourcing of heater fabrication of a NiCr heater with polyamide in place of silicone. Final heater was right-first-time and was implemented in a newer iteration of the same product.

2.3.2. Insights

The case demonstrated the impact of rapid iterations and early testing for PCB design, providing a right-first-time outsourced FPC. Quick probing prototypes, designed for prototyping and not a product, revealed a viable path fast, and rapid iteration through developed tooling, was detrimental to efficiently find a satisfactory solution. While one could simulate the perfect NiCr layout, physically assembling it revealed the physical limitation. This illustrates the potential power of desktop manufacturing in prototyping PCBs.

3. Recommendations for rapid prototyping of PCBs

Rapid prototyping methods applied to PCB prototyping arguably yielded effective prototyping processes. Through seeking rapid iteration, low-fidelity prototypes, and early testing of concepts the cases indicated substantial benefit from applying rapid prototyping methodologies. From the insights acquired through the cases presented we have sought to formalize recommendations for rapid prototyping of PCBs.

3.1. Increase fidelity deliberately

Move to PCBs when higher resolution or fidelity are genuinely required. For example, when finer pitch, repeatable geometry, mechanical robustness, material requirements, etc. meets constraints that hand builds cannot meet. Before outsourcing, consider “hacking” prototypes, using in-house or hybrid

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

processes (Milling, Laser-cutting, Additive production), to answer questions. Stay aware of their capabilities and limitations though, to reduce the risk of spending unnecessary time on PCB work. Time spent on designing and producing PCB prototypes through manufacturing or hacking is illustrated in Figure 5. Another consideration is if you are dealing with sensitive systems, you might have time saved from troubleshooting for example a loose connection or with sensitive components prone to data loss. This is only if fidelity is rather high. This recommendation helps avoid the PCB limitation of skill overhead and time, by avoiding PCBs when not strictly necessary.

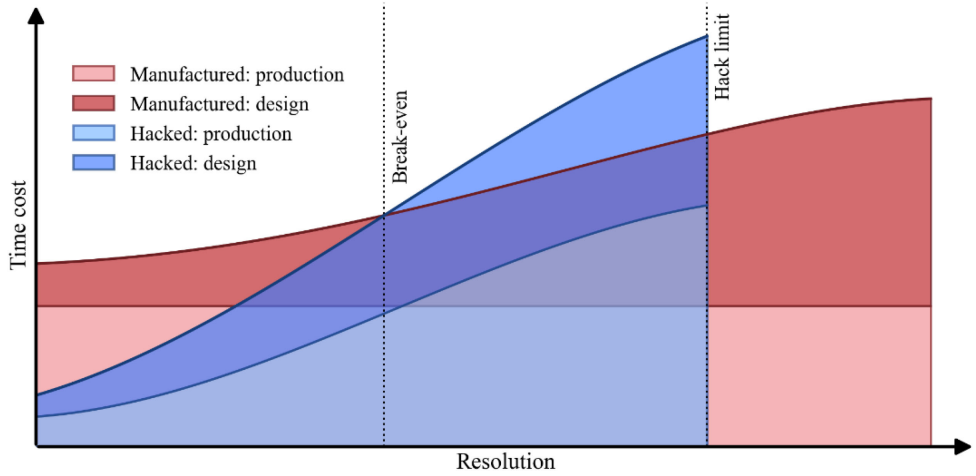


Figure 5. Representative graph of prototype resolution vs time cost of design and production for manufactured and hacked PCBs. There is a break-even point where the resolution required is more effectively manufactured, and a limit of obtainable resolution while hacking

3.2. Design for prototyping

Treat early PCBs as prototypes, as instruments for learning, not products. Keep clear design questions in mind, a prototype should be an answer to this question. Down-scale and simplify to reduce risk and decrease design time, only include what is strictly necessary to answer the question at hand. Modularize: keep low-fidelity concepts as sub-assemblies of different mediums to allow for iterating towards a higher fidelity. Favour hand-solderable components packages, use solder jumpers, and design accessible test points, solder pads and through-holes to enable troubleshooting and manual rework, reducing the risk of unusable prototypes.

3.3. Iterate incrementally

Change one major variable per iteration and keep the others constant. Extensive iteration combining multiple unproven ideas is not only labour intensive in terms of designing the iteration but also induces the risk of unsuccessful aspects of a prototype leaving others design questions unanswered. Instead, separate design variables to carefully address individual questions. Use panelisation to bundle several small experiments or iterations into one PCB to reduce design time and utilise manufacturing time for multiple experiments. Panelisation entails a singular PCB with separate circuit designs and can also be used to explore several design options concurrently. Identify the design questions that do not require PCB manufacturing to instead answer them in alternative ways, as described in 3.1.

3.4. Parallelise prototyping

Exploit inevitable fabrication lead times by advancing other design aspects in parallel. Develop firmware, data processing, or revise other sub-system designs to spend the waiting time progressing the process and to maximise learnings from prototypes.

3.5. Prototype and test early

Get data early, even from low-fidelity embodiments. Early data, either qualitative or quantitative can surface core behaviours or constraints long before polished hardware is made. Especially physical

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

aspects of a PCB design, early tests can reveal issues that no simulation or specification could have uncovered as quickly. Uncovering these unknown unknowns early might reduce the risk of costly iterations.

4. Discussion

While the established PCB design workflows likely provide necessary support in the development of complex circuitry, we argue that these do not apply to the development of low-complexity PCBs, particularly when PCBs are used outside of their standard use. There is a lack of a guidelines enabling effective prototyping of PCBs where the solution space is ambiguous and requirements are not fixed. This need will only grow with the continued increase in the interdisciplinarity in product development. We call for an extension or rework of design workflows to encourage and enable rapid prototyping of and with PCBs, removing discipline separation and intensive reviews, deploying early iteration and testing. A rapid PCB prototyping workflow should more closely resemble an unstructured process, moving decisively between steps as the designers deem appropriate, along the lines of Figure 6. It is an attempt to adapt the workflow presented in Figure 1 into one more fitting for rapid prototyping PCBs. Notably, we have removed steps we deem not applicable in this context, removed the linear nature in favour of a less structured approach, "tore down that wall" separating design and CAD departments, and added the step of prototyping with available materials. We believe a workflow along these lines might benefit less defined PCB design processes.

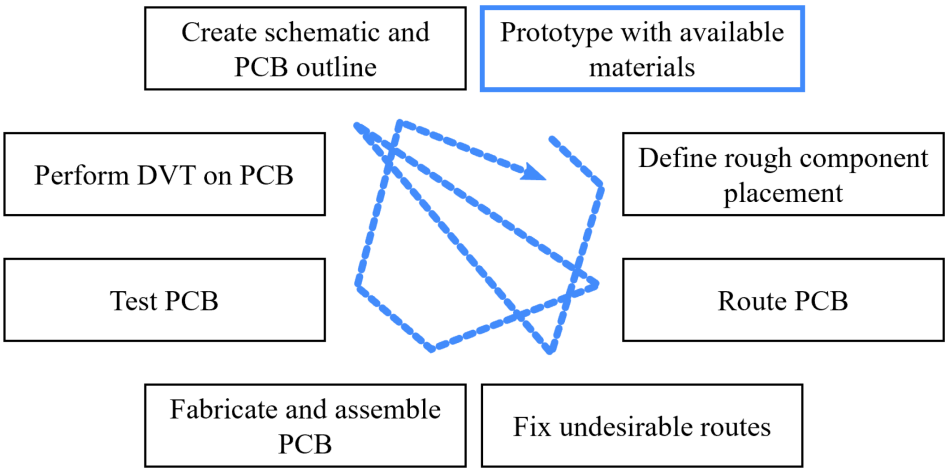


Figure 6. Proposed process. Throughout the iterative product development process, the developer will move between the different tasks as they earn insights and learn what is needed for the problems at hand. The lines are arbitrary and represent that the process should be unstructured

Generalizability might be limited, as case studies entailed untraditional use-cases for PCBs. These open solution spaces called for exploration in the early phase of development, but the methodology might not have proven as effective in later less open-ended stages. As mentioned, complicated and intricate circuitry where methods are well established will likely not benefit from integrating a rapid prototyping mindset, but we argue that current workflows do not provide the necessary support for medium to low-complication PCB designs and believe adapting these insights would prove helpful. This should be explored in future research.

Skill needs to be addressed to fully enable PCBs as a rapid prototyping tool. Having more experienced users around allows designers to make "skill-jumps" (Vestad et al., 2019), enabling more efficient acquiring of the skill of PCB design. Increased interdisciplinary collaboration in educational institutions and commercial companies alike could further support this effect.

Rapid prototyping of PCBs could benefit from improved in-house production capabilities. Several crowdsourced commercial desktop PCB printers have been introduced in the last decade (Espera et al., 2019). The ability to produce higher resolution PCBs in-house would naturally yield shorter iteration cycles. In-house production capabilities would give the additional benefit of independence from supply chain. In research, both additive and subtractive in-house PCB production has been explored (Bin Islam et al., 2024; Espera et al., 2019; Toprak & Boynuegri, 2020; Yan et al., 2022), however there does not

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

seem to exist a widely adopted dominant solution. The diversity and multitude of proposed solutions to address desktop PCB production certainly indicates a want from the community for these capabilities. Concludingly, though evidently solutions exist, it is the hope of the authors that these developments will make PCB production increasingly accessible, enabling even more prototyping with PCBs in the future.

References

Berschik, M. C., & Krause, D. (2024). Towards Model-Based Modular Product Families in the Context of Systems Engineering. *DS 130: Proceedings of NordDesign 2024, Reykjavik, Iceland, 12th - 14th August 2024*, 71–78. https://doi.org/10.35199/NORDDESIGN2024.8

Bilal, M., Eynard, B., & Guérineau, J. (2025). A community-driven database for the dynamic representation of approaches, processes, methods, and tools for multidisciplinary product development. *Proceedings of the Design Society*, 5, 1–10. https://doi.org/10.1017/pds.2025.10014

Bin Islam, M. S., Shaikhul Hasib, M., Shakhawat Hossen, M., Tareq Aziz, S. M., & Rokunuzzaman, M. (2024). Design and Implementation of a CoreXY CNC Milling Machine for PCB Manufacturing Application (SSRN Scholarly Paper No. 4860937). *Social Science Research Network*. https://doi.org/10.2139/ssrn.4860937

Canal Marques, A., Cabrera, J.-M., & De Fraga Malfatti, C. (2013). Printed circuit boards: A review on the perspective of sustainability. *Journal of Environmental Management*, 131, 298–306. https://doi.org/10.1016/j.jenvman.2013.10.003

Dow, S. P., Heddleston, K., & Klemmer, S. R. (2009). The efficacy of prototyping under time constraints. *Proceedings of the Seventh ACM Conference on Creativity and Cognition*, 165–174. https://doi.org/10.1145/1640233.1640260

Eisenhardt, K. M. (1989). Building Theories from Case Study Research. *Academy of Management Review*, 14(4), 532–550. https://doi.org/10.5465/amr.1989.4308385

Espera, A., Dizon, J. R., Chen, Q., & Advincula, R. (2019). 3D-printing and advanced manufacturing for electronics. *Progress in Additive Manufacturing*, 4. https://doi.org/10.1007/s40964-019-00077-7

Goti, A. B. (2025a). 3D-Printed Multi-Layer PCBs: Evaluating the Structural Integrity and Electromagnetic Compatibility of Additively Manufactured Circuits. *International Journal of Engineering and Computer Science*, 14(06), 27258–27270. https://doi.org/10.18535/ijecs.v14i06.5158

Goti, A. B. (2025b). Hybrid Additives-Subtractive Manufacturing of Multi-Layer PCBs Using Laser Direct Structuring (LDS) and Inkjet printing. *International Journal of Scientific Research and Management (IJSRM)*, 13, 2242–2253. https://doi.org/10.18535/ijsrm/v13i06.ec04

Han, C., Zhang, C., Tang, W., Li, X., & Wang, Z. L. (2015). High power triboelectric nanogenerator based on printed circuit board (PCB) technology. *Nano Research*, 8(3), 722–730. https://doi.org/10.1007/s12274-014-0555-3

Houde, S., & Hill, C. (1997). Chapter 16—What do Prototypes Prototype? In M. G. Helander, T. K. Landauer, & P. V. Prabhu (Eds), *Handbook of Human-Computer Interaction* (Second Edition) (pp. 367–381). North-Holland. https://doi.org/10.1016/B978-044481862-1.50082-0

IPC-2221A: Generic Standard on Printed Board Design (No. IPC-2221A). (2003). IPC.

Jansson, D. G., & Smith, S. M. (1991). Design fixation. *Design Studies*, 12(1), 3–11. https://doi.org/10.1016/0142-694X(91)90003-F

Jeong, H., Tentzeris, M. M., & Lim, S. (2018). Frequency-Tunable Electromagnetic Absorber by Mechanically Controlling Substrate Thickness. *International Journal of Antennas and Propagation*, 2018, 1–7. https://doi.org/10.1155/2018/1963051

Kriesi, C. (2018). Wayfaring in the Biomedical Sector: A Call for Re-Introducing the Toolmaker [Doctoral thesis, NTNU]. https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/2584988

Kriesi, C., Blindheim, J., Bjelland, Ø., & Steinert, M. (2016). Creating Dynamic Requirements through Iteratively Prototyping Critical Functionalities. *Procedia CIRP*, 50, 790–795. https://doi.org/10.1016/j.procir.2016.04.122

Kriesi, C., Steinert, M., Marmaras, A., Danzer, C., Meskenaite, V., & Kurtcuoglu, V. (2019). Integrated Flow Chamber System for Live Cell Microscopy. *Frontiers in Bioengineering and Biotechnology*, 7. https://doi.org/10.3389/fbioe.2019.00091

LaDou, J. (2006). Printed circuit board industry. *International Journal of Hygiene and Environmental Health*, 209(3), 211–219. https://doi.org/10.1016/j.ijheh.2006.02.001

Leifer, L. J., & Steinert, M. (2011). Dancing with ambiguity: Causality behavior, design thinking, and triple-loop-learning. *Information Knowledge Systems Management*, 10(1–4), 151–173. https://doi.org/10.3233/IKS-2012-0191

Low, Z., Cheong, J. H., & Law, C. (2005). Low-cost PCB antennas for UWB applications. *Antennas and Wireless Propagation Letters*, IEEE, 4, 237–239. https://doi.org/10.1109/LAWP.2005.852577

Moreira, F. T. C., Ferreira, M. J. M. S., Puga, J. R. T., & Sales, M. G. F. (2016). Screen-printed electrode produced by printed-circuit board technology. Application to cancer biomarker detection by means of plastic

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press

antibody as sensing material. *Sensors and Actuators B: Chemical*, 223, 927–935. https://doi.org/10.1016/j.snb.2015.09.157
Moschou, D., & Tserepi, A. (2017). The lab-on-PCB approach: Tackling the μTAS commercial upscaling bottleneck. *Lab on a Chip*, 17(8), 1388–1405. https://doi.org/10.1039/C7LC00121E
Narumi, K., Shi, X., Hodges, S., Kawahara, Y., Shimizu, S., & Asami, T. (2015). Circuit Eraser: A Tool for Iterative Design with Conductive Ink. *Proceedings of the 33rd Annual ACM Conference Extended Abstracts on Human Factors in Computing Systems*, 2307–2312. https://doi.org/10.1145/2702613.2732876
Perdigones, F. (2021). Lab-on-PCB and Flow Driving: A Critical Review. *Micromachines*, 12(2), 175. https://doi.org/10.3390/mi12020175
Research and Markets. (2025). Printed Circuit Board Market Size, Share & Forecast to 2033. https://www.researchandmarkets.com/reports/5732732/printed-circuit-board-market-report-type
Ritchey, L. W. (2016). The PCB Design Process. In Coombs, Clyde F., Jr. & Holden, Happy T. (Eds), *Printed Circuits Handbook* (7th ed., pp. 385–394). McGraw-Hill Education.
Rodgers, T. (2016). Design for Manufacturability. In Coombs, Clyde F., Jr. & Holden, Happy T. (Eds), *Printed Circuits Handbook* (7th ed., pp. 43–62). McGraw-Hill Education.
Shamkhalichenar, H., Bueche, C. J., & Choi, J.-W. (2020). Printed Circuit Board (PCB) Technology for Electrochemical Sensors and Sensing Platforms. *Biosensors*, 10(11), 159. https://doi.org/10.3390/bios10110159
Sjöman, H. (2019). Insights for Prototyping for the Industrial Internet-Iterative Prototyping Loops of Connected Sensor Platforms [Doctoral thesis, NTNU]. https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/2611194
Toprak, A., & Boynuegri, A. R. (2020). Printed Circuit Board Rapid Prototyping with Three-Dimensional Printer. *2020 4th International Symposium on Multidisciplinary Studies and Innovative Technologies (ISMSIT)*, 1–7. https://doi.org/10.1109/ISMSIT50672.2020.9254998
Ulrich, K. T., & Eppinger, S. D. (2016). *Product design and development*. McGraw-Hill. https://thuvienso.hoasen.edu.vn/handle/123456789/9147
Vestad, H., Kriesi, C., Slåttsveen, K., & Steinert, M. (2019). Observations on the Effects of Skill Transfer through Experience Sharing and In-Person Communication. *Proceedings of the Design Society: International Conference on Engineering Design*, 1(1), 199–208. https://doi.org/10.1017/dsi.2019.23
Vestad, H. N. (2018). Artificial Lateral Line Sensors-Developing Surface Sensors for Flow Detection around Adaptive Hydrofoils [Master's Thesis, NTNU]. https://ntnuopen.ntnu.no/ntnu-xmlui/handle/11250/2558746
Vestad, H., & Steinert, M. (2019). Piezoresistive Chopped Carbon Fiber Rubber Silicone Sensors for Shedding Frequency Detection in Alternating Vortex Streets. *2019 IEEE SENSORS*, 1–4. https://doi.org/10.1109/SENSORS43011.2019.8956632
Vu, Q. T., Nguyen, L. L., & Pham, V. N. (2024). Compact lab-on-printed circuit board (PCB) for free-surfactant silver nanomaterial synthesis. *Chemical Engineering and Processing - Process Intensification*, 203, 109918. https://doi.org/10.1016/j.cep.2024.109918
Yan, Z. (2024). Sustainable in-house PCB prototyping. *The 37th Annual ACM Symposium on User Interface Software and Technology*, 1–5. https://doi.org/10.1145/3672539.3686710
Yan, Z., Sathya, A., Yusuf, S., Lien, J.-M., & Peng, H. (2022). Fibercuit: Prototyping High-Resolution Flexible and Kirigami Circuits with a Fiber Laser Engraver. *Proceedings of the 35th Annual ACM Symposium on User Interface Software and Technology*, 1–13. https://doi.org/10.1145/3526113.3545652
Youmans, R. J. (2011). The effects of physical prototyping and group work on the reduction of design fixation. *Design Studies*, 32(2), 115–138. https://doi.org/10.1016/j.destud.2010.08.001
Zivanovic, S., Popovic, M., Nikola, V., Pjevic, M., & Slavković, N. (2020). An Overview of Rapid Prototyping Technologies using Subtractive, Additive and Formative Processes. *FME Transactions*, 48, 246–253. https://doi.org/10.5937/fmet2001246Z

https://doi.org/10.1017/pds.2026.10438 Published online by Cambridge University Press